

Clinical Risk Assessment Report: Patient 61

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 15.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.69x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.8%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.63 [Avg (66th %ile)] (+1.8%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.84 [Top 19%] (+1.6%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.6 [Bottom 18%] (+1.3%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.18 [Avg (54th %ile)] (-1.8%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

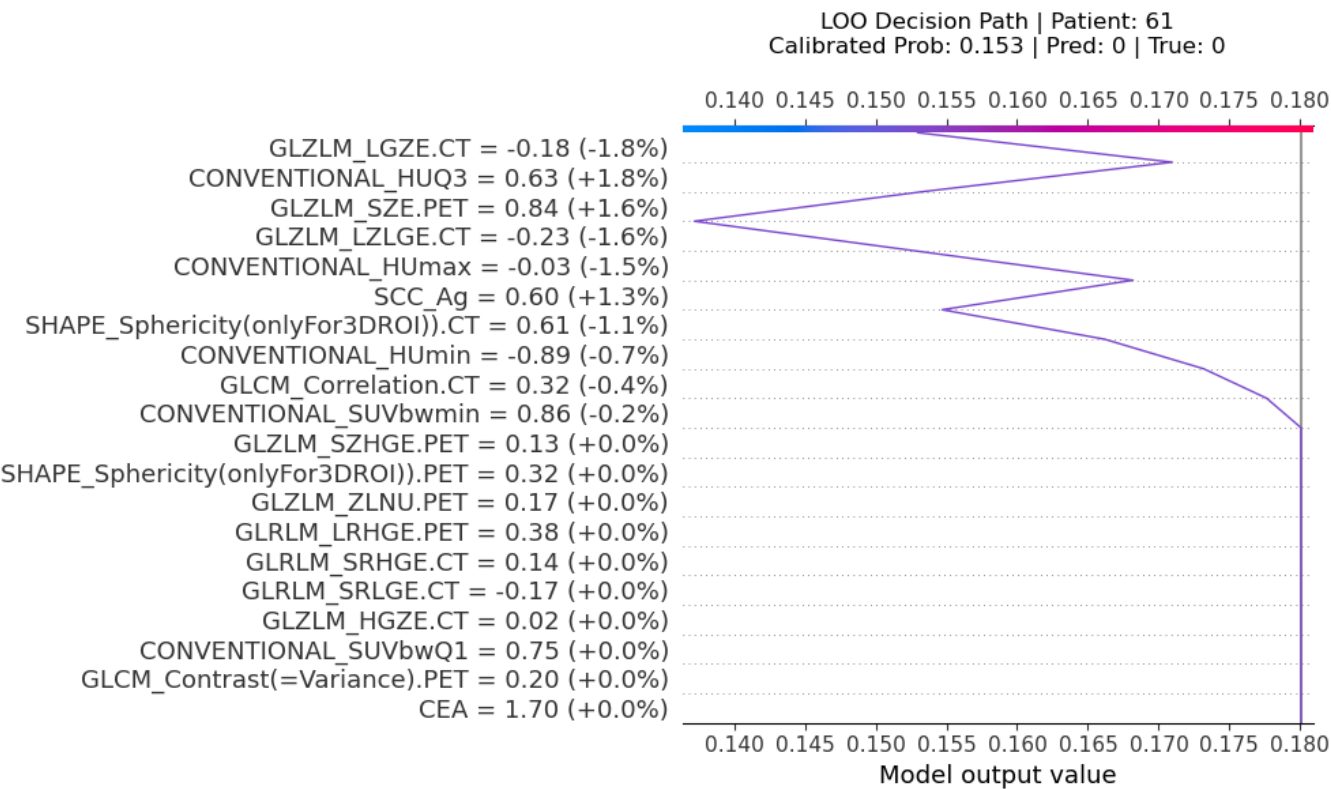
Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.23 [Bottom <1%] (-1.6%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.03 [Avg (44th %ile)] (-1.5%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.61 [Avg (73th %ile)] (-1.1%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.89 [Bottom 17%] (-0.7%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

